

34229/34249 - Cyber Technology/Communication Technologies

Introduction to (Photonic Related) Information Resources

Henning K. Kristensen from DTU Library – hkkkr@dtu.dk

Agenda

- **What is the point?**
 - Why to search
 - How to search
- **Tools provided by DTU Library**
 - DTU Findit
 - Web of Science & Scopus
 - Ebook collections and links to more...
- **Short pitch for subject specific tools**
 - IEEE Explorer & SPIE Digital Library
 - Derwent Innovations Index
- **Reference management (Mendeley)**



*Questions &
discussions are more
than welcome as we
go along 😊*

What is the point?

- We want to conduct a systematic, scientific and comprehensive literature search in all sources relevant to our field of study.
- We want our search to be transparent and reproducible -> similar to a lab protocol.
- We want to take full advantage of the curated, quality assured, human indexed and expensive bibliographic databases provided by DTU Library (such as Scopus, IEEE Explorer, Derwent II etc..)
- Why are we not using fancy AI or Google Scholar?

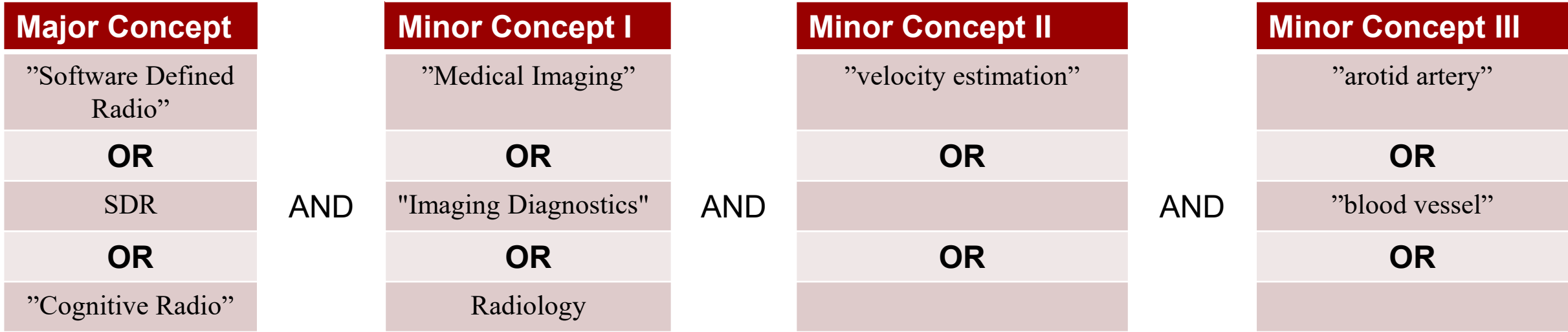
Things to consider in the planning phase

- Identify information sources (books, articles (review or original research), reports, patents, facts..)
- Find/locate the information (via library resources, web, friends/colleagues...)
- Get access to information (DTU Findit, other library tools, Google, friends/colleagues..)
- Evaluate the findings (reading snippets/abstracts, choosing, being critical..)
- Manage the findings (export results to reference management, documentation, citations..)



The Matrix

Problem Statement: ..something about Medical imaging using Software Defined Radio?



You

Dear chat gpt. Can you please help me with synonyms for Software Defined Radio?

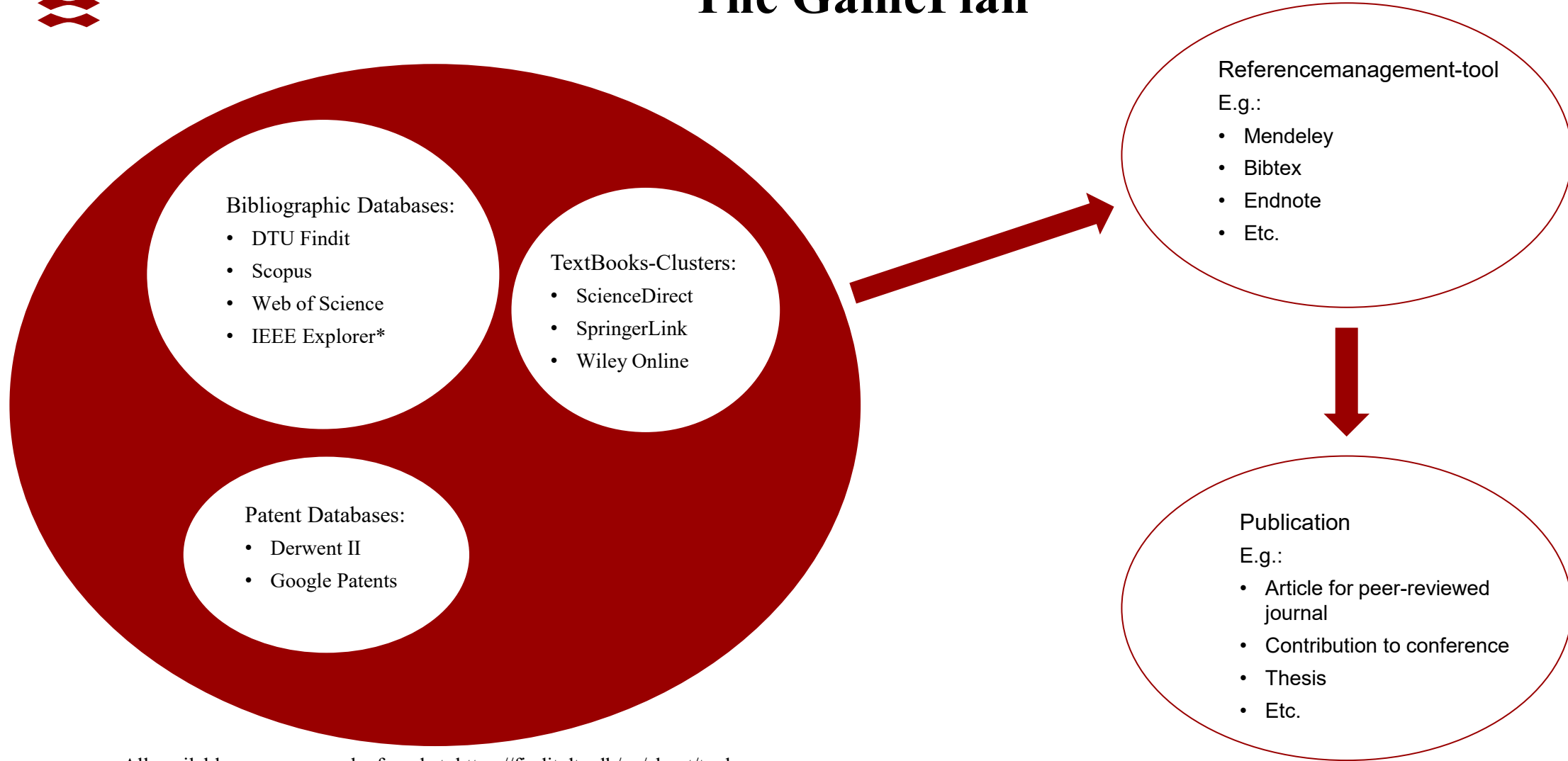
ChatGPT

Certainly! Here are some synonyms and related terms for Software Defined Radio (SDR):

- Software Radio
- Software-Defined Wireless
- Cognitive Radio
- Digital Radio
- Software-Defined Receiver
- Flexible Radio
- Reconfigurable Radio
- Software-Defined Transceiver
- Virtual Radio
- Programmable Radio

You can use AI and/or Wiki when appropriate

The GamePlan



All available resources can be found at: <https://findit.dtu.dk/en/about/tools>

*Journal articles and conference contributions from IEEE are included in Scopus and DTU Findit, but not the IEEE standards.

DTU Findit: the entry point to everything

Gives you access to millions of scientific articles, books, theses, reports all through one single entry point –

Mostly in english, but you can search by use of danish keywords (incl. danish articles and book titles)

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- author:(Van Deursen) format:article title:(Improving digital skills for the use of online public information and services)

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- (digital competences OR digital skills) AND (civil servants OR society)
- ("type 2 diabetes" OR "type 1 diabetes") AND (treatment OR cure) TYPE: Ebook
- (tagging OR leading) AND indicator AND ecodesign
- (ecodesign OR "ecological design") AND (implementing OR operation OR realization) AND denmark
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11

1

Other search tools

New tool

A B C D E F G H I J K L M N O P R S T U V W Z

A

ACM Digital Library

Provided by DTU Library. ACM Digital Library covers IT and computer science. It contains bibliographical references, conference proceedings, reviews as well as journal and newsletter articles published by the Association for Computing Machinery and affiliated organisations (such as ACL, SIGCAS, SIGGRAPH). Records go back to the 1950s. Content is integrated with and searchable in DTU Findit.

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AES E-Library

Provided by DTU Library. AES E-Library covers audio technology. The tool contains a collection of articles and documents published by the Audio Engineering Society, including convention papers, conference papers, engineering briefs, technical documents, and standards. Records go back to 1935.

Electrotechnology · Medicine and Medical Technology

ASM Handbooks

Provided by DTU Library. ASM Handbooks covers information on ferrous and non-ferrous metals and materials technology. Maintained by The American Society for Metals (ASM), the database contains 30 ASM Handbook volumes, comprising more than 25,000 pages of articles, illustrations, graphs, specifications, performance charts, and processing guidelines. Updated quarterly.

Chemistry · Construction and Mechanics · Physics · Materials

ASTM Compass

Provided by DTU Library. ASTM Compass provides access to standards, journals, manuals, monographs, research reports, and technical publications published by The American Society for Testing and Materials (ASTM). Content is integrated with and searchable in DTU Findit.

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Academic Search Elite

Provided by DTU Library. Academic Search Elite covers a large number of subjects within Engineering, Technology and Natural Sciences. It mainly contains e-books and journal articles. Full-text records go back to 1985.

Information Technology · Construction and Mechanics · Production and Management · Transport and Logistics

AccessEngineering

Provided by DTU Library. AccessEngineering is an online engineering reference tools containing full-text handbooks, as well as instructional, faculty made videos, calculators, interactive tables and charts.

Space Research · Chemistry · Construction and Mechanics · Energy · Electrotechnology · Physics · Earth Sciences · Materials

AccessScience

Provided by DTU Library. AccessScience covers all major disciplines within science and technology. In addition to articles and research reviews, the database contains over 9000 encyclopedia articles. It provides a gateway to scientific knowledge, covering both basic introductory material as well as extended article entries and live journal citations.

Altinet

Provided by DTU Library. Altinet is a Danish news site that offers information on the latest political developments related to Danish (and European) politics including parliamentary debates, features, interviews, letters to the editor. The website also

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Search within

Article title, Abstract, Keywords

▼

Search documents *

"Software Defined Radio" OR sdr OR "Cognitive Radio"

×



Koncept I

AND ▼

Search within

Article title, Abstract, Keywords

▼

Search documents

"medical imag*" OR "Imaging Diagnostics" OR Radiology

×



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Scopus – bibliographic database. How to use it?



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Article title, Abstract, Keywords

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"Software Defined Radio" OR sdr OR "Cognitive Radio"

AND

Search within
Article title, Abstract, Keywords

Search documents
"medical imag*" OR "Imaging Diagnostics" OR radiology

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☐ 1	Conference Paper Deep Learning based Spectrum predictions for Cognitive Healthcare system	Kansal, P.	2023 14th International Conference on Computing Communication and Networking Technologies, ICCCNT 2023	2023	0

Show abstract

DTU Findit

Related documents

Available via DTU Findit



IEEE Transactions on Microwave Theory and Techniques • Volume 64, Issue 2, Pages 643 - 652 • 1 February 2016 •
Article number 7378332

Software-defined radar for medical imaging

Marimuthu, Jayaseelan ; Bialkowski, Konstanty S. ;

Abbosh, Amin M.

Abstract

A low-cost reconfigurable microwave transceiver using software-defined radar is proposed for medical imaging. The device, which uses generic software-defined radio (SDR) technology, paves the way to replace the costly and bulky vector network analyzer currently used in the research of microwave-based medical imaging systems. In this paper, calibration techniques are presented to enable the use of SDR technology in a biomedical imaging system. With the aid of an RF circulator, a virtual 1-GHz-wide pulse is generated by coherently adding multiple frequency spectrums together. To verify the proposed system for medical imaging, experiments are conducted using a circular scanning system and directional antenna. The system successfully detects small targets embedded in a liquid emulating the average properties of different human tissues. © 2016 IEEE.

Author keywords

Microwave transceiver; Software-defined radar (SDRadar); Terms-Microwave imaging

Indexed keywords

Engineering controlled terms

Directive antennas; Electric network analyzers; Imaging systems; Microwaves; Radar; Radar imaging; Radio transceivers; Scanning antennas; Software radio; Transceivers

Engineering uncontrolled terms

Biomedical imaging; Calibration techniques; Directional Antenna; Microwave imaging; Microwave transceivers; Multiple frequency; Software defined radar; Vector network analyzers

Engineering main heading

Medical imaging

Indexed keywords – the true human value add, that is what we pay for: ←

DTU Library provide access to several E-book collections

Photonic Devices

Of all photonic devices, the photonic crystal cavity provides the best confinement in both space in time, which is promising for low power optical links as the energy consumption of a photonic device is proportional to the product of the photon loss rate and the volume.

From: [Optical Interconnects for Data Centers](#), 2017

Related terms:

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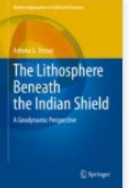
Photonics in *Earth and Planetary Sciences*
Photonics in *Engineering*
Photonics in *Physics and Astronomy*
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Si **Photonics** in *Engineering*
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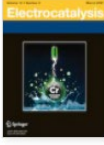
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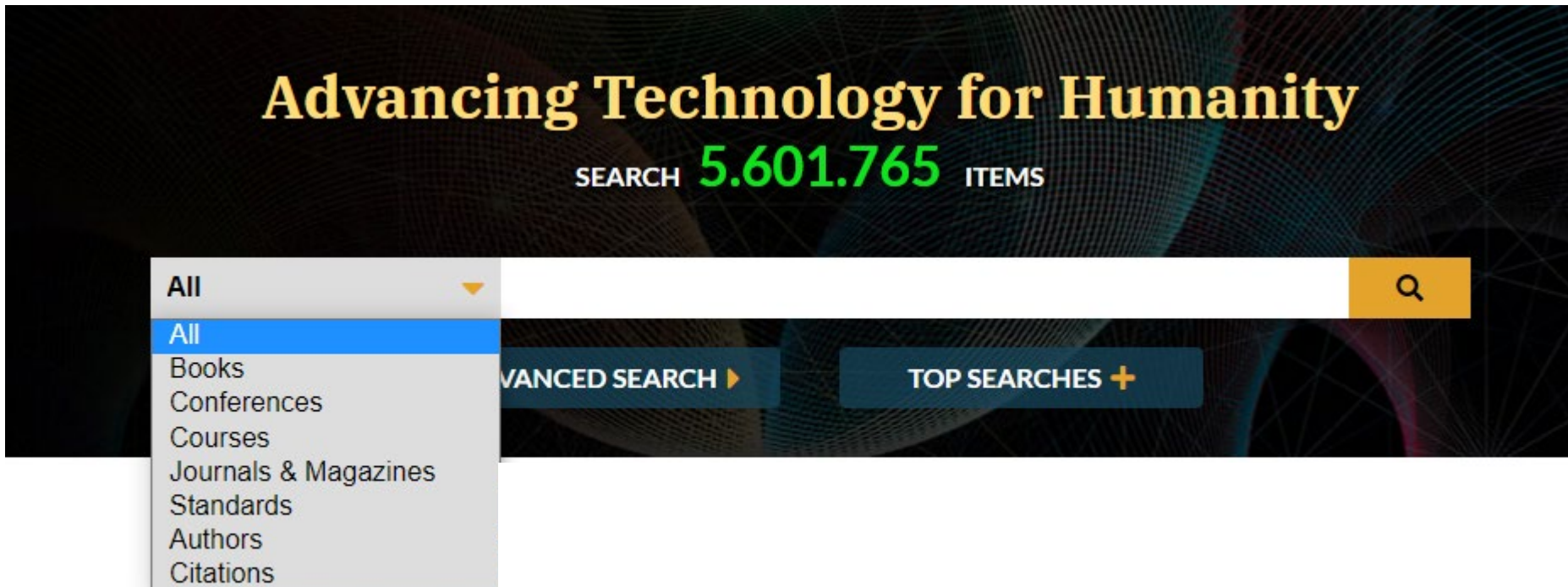
   

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Authors



henrik

wessing



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☐ Novel scheme for packet forwarding without header modifications in opt

H. Wessing; H. Christiansen; T. Fjelde; L. Dittmann

Journal of Lightwave Technology

Year: 2002 | Volume: 20, Issue: 8 | Journal Article | Publisher: IEEE

Cited by: Papers (18)

► Abstract

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Front Matter: Volume 7314

Proc. SPIE. 7314, Photonics in the Transportation Industry: Auto to Aerospace II

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Journal Article | 28 August 2014  Open Access

Review of diverse optical fibers used in biomedical research and clinical practice

Gerd Keiser, Fei Xiong, Ying Cui, Perry Ping Shum

Journal of Biomedical Optics Vol. 19, Issue 8 (Aug 2014)

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Journal Article | 25 June 2020  Open Access

Fiber Bragg grating sensors for monitoring of physical parameters: a comprehensive

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"fiber bragg gratings"



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Searches title and abstract.

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Enzym* water consumption

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- ☐ Univ Kunming Sci Technology 9

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>

- ☐ 1 Optical fiber Raman laser using optical fiber Bragg gratings and remote sensor using the same, particularly for realizing remote sensor for over 25km deep sea and underground

KR2005099744-A, KR669536-B1

Inventor(s) : [LEE J H](#); [HAN Y G](#); (...); [KIM S H](#)Assignee(s) : [KOREA INST SCI & TECHNOLOGY](#)

Derwent Primary Accession Number :

2006-666096

...

3

[Citing Patents](#)

- ☐ 2 Parallel optical fiber grating, has four fiber Bragg gratings whose four intercept points are placed on circular cross section of cylinder, where modulation directions of two adjacent fiber Bragg gratings are different by specific degrees

CN109445022-A; CN209265001-U; WO2020140253-A1

Inventor(s) : [WANG Y](#); [LIAO C](#) and [LI Z](#)Assignee(s) : [UNIV SHENZHEN](#)

Derwent Primary Accession Number :

2019-25197R

Quantum communication system using code division multiple access network, has three optical fiber elements as basic building blocks of working system, such as fiber Bragg gratings, circulators, and electro-optic modulators

Patent Number: IN202011034133-A

Inventor: SHARMA V

Patent Assignee:

SHARMA V(SHAR-Individual)

Derwent Primary Accession Number: 2022-343496

Indexed: 2022-03-27

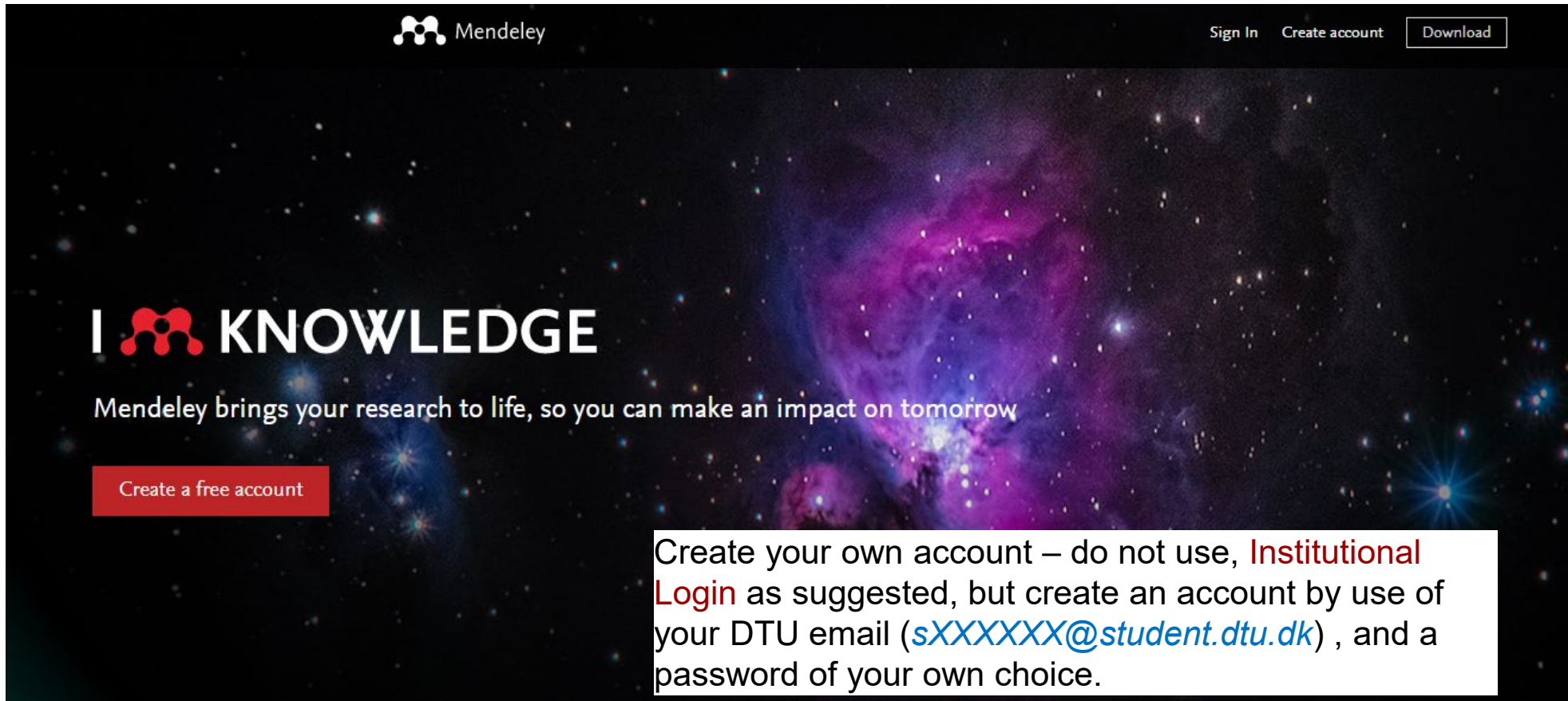
Abstract:

NOVELTY - The system has three optical fiber units as basic building blocks of a working system. Fiber Bragg gratings (FBG) drop an optical signal by circulators using a dense wavelength division multiplexing (DWDM) technique. The circulators with a center wavelength of 1310 nm, 1550 nm, and 1064 nm are non-reciprocal optical devices that redirect incoming optical signals to successive output ports. The fiber Bragg grating allows multiple of the incoming signals to pass without altering their properties, when a special frequency band is reflected.

USE - Quantum communication system using code division multiple access network for quantum communication protocols such as quantum teleportation, quantum key distribution, and other quantum information networks (Nielsen and Chuang 2001).

ADVANTAGE - The method enables realizing quantum communication with significant reduced information loss using error correction methods, thus enhancing capacity of a quantum channel by correcting errors at a receiver end after prudent encoding in a photon-environment. The method allows concurrent but uncoordinated quantum communication among multiple parties

Introduction to Mendeley & Reference management



The screenshot shows the Mendeley website homepage. At the top, there is a navigation bar with the Mendeley logo, "Sign In", "Create account", and "Download" buttons. The main content area features a large, vibrant image of a nebula in space. Overlaid on this image is the text "I KNOWLEDGE" in large, white, bold letters, with the Mendeley logo replacing the letter "I". Below this, a smaller line of text reads "Mendeley brings your research to life, so you can make an impact on tomorrow". A red button with the text "Create a free account" is positioned on the left. On the right, a white text box contains instructions for account creation.

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- is a very useful tool when it comes to collecting references, annotate and organise used references, share references and keep track of useful information for **Project assignments or other studies**

.....this is a very Quick introduction that includes getting access and a few pointers on "how to get the best out of .."



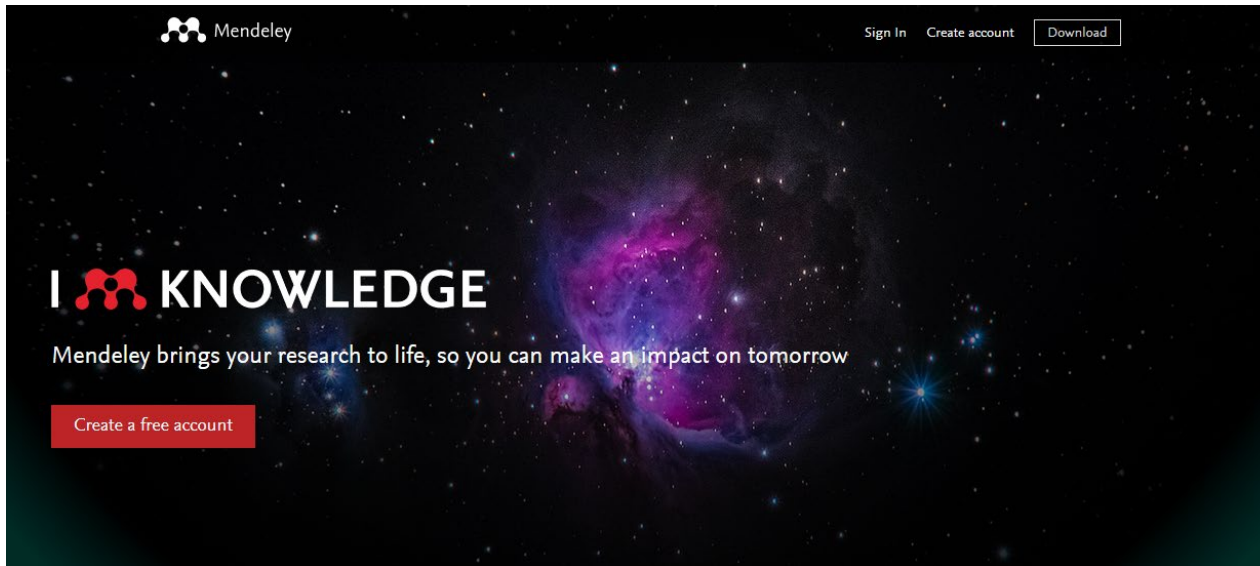
Reference Management by use of Mendeley

**Mendeley Institutional Edition via DTU Library
includes a Write'N'Cite plug-in to Word/Libre Office**

- Mendeley provides an **empty database, for you to built**
- **Getting started** (create an account at www.mendeley.com)
- **Import references** (from DTU Findit, Web of Science, Scifinder & even Google Scholar)
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- **Citations & Bibliographies are** easily created in Word (Mendeley Word Plug-in OR Mendeley Cite app)

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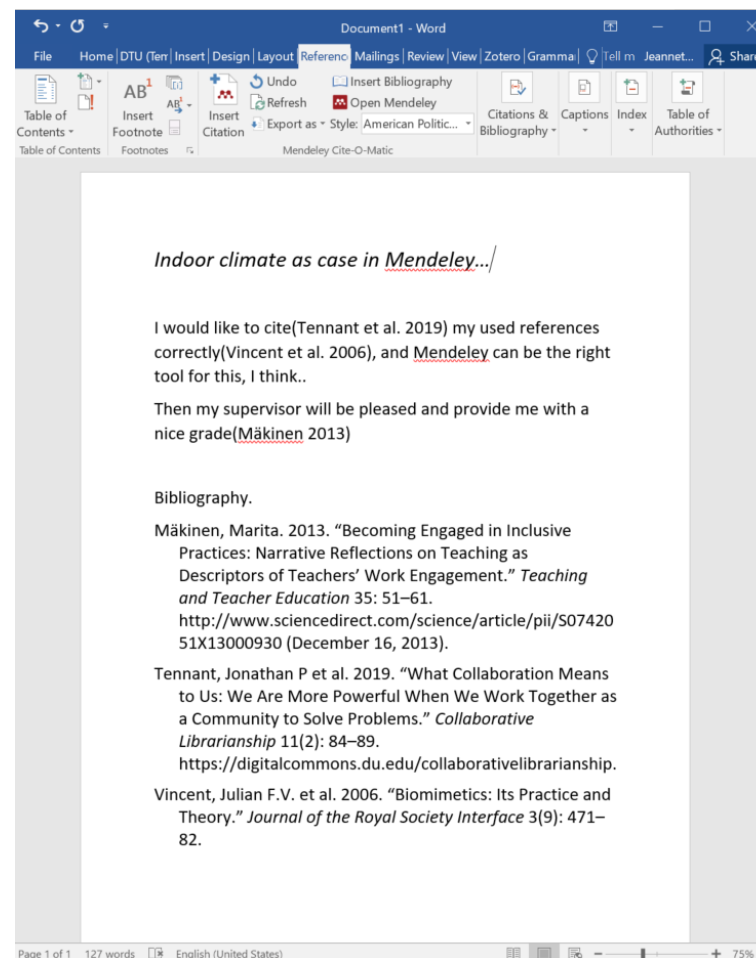
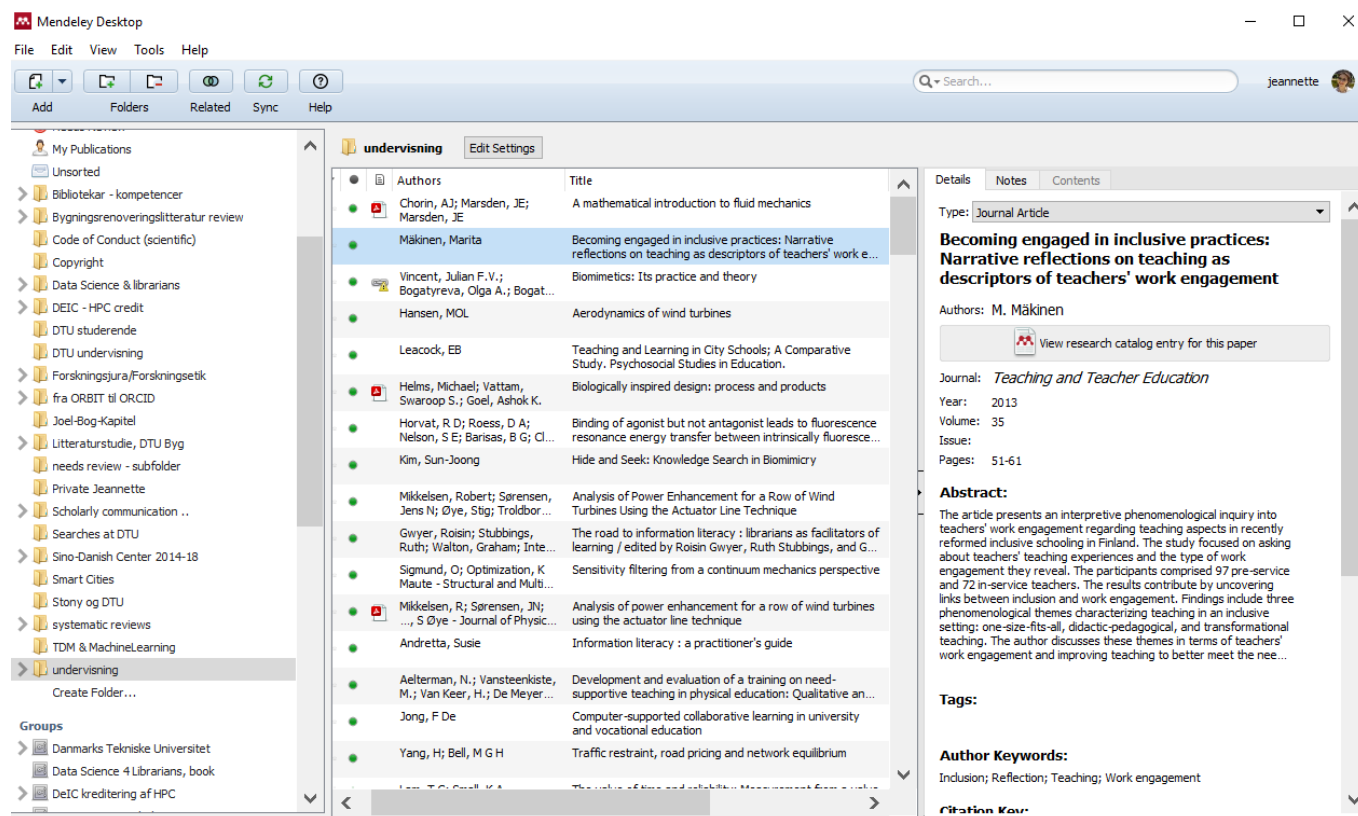
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- reference management with Mendeley

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- Working with your references in Word or exporting to BibTEX
- How you share references and create private groups
- Tips for using Mendeley for non-Word users (Latex/ Overleaf)

The webinar is just an introduction; we expect that you have made an account with Mendeley beforehand and installed the necessary plugins - see links below.

For more advanced use of Mendeley, please contact us via mail or [book an appointment with a librarian](#).

<https://www.bibliotek.dtu.dk/en/calendar/mendeley-20032025?id=b390acf3-4a5a-4984-bb92-7ff1968fae4a>

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